Input

Target RTL

module dct_top
...
endmodule

+

Approximate Unit Library

+ × + ...

Graph

Co-Evolution

Candidate Search
NSGA-III

Surrogate Models

GNNs

Area / Power
Latency
SSIM
max
Path-Aware readout

Surrogate Prediction

Models Update

Logic Synthesis Accuracy Evaluation

Pareto-relevant Selection

Local Calibration
Latency only

Reference Update

Candidate States

- Unevaluated
- $\hat{y}$ Predicted PPA / SSIM
- $\tilde{y}$ Latency-calibrated
- y Measured

y = (Area, Power, Latency, SSIM)

Output

Pareto Front

Representative Designs

...

Final DSE

Final Search

Selected Designs

Logic Synthesis Accuracy Evaluation

BSDS Images

Results